

Adversarial Autoencoders with Diffusion-Based Latent Denoising

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Abstract

Adversarial Autoencoders (AAEs) are a powerful class of generative models that map input data into a structured latent space, regularized via adversarial training. However, latent vectors produced by the encoder may include noise that degrades the quality of reconstructed or generated samples. This report proposes a novel Hybrid AAE architecture that integrates a denoising Diffusion model between the encoder and decoder to mitigate the impact of latent noise. Both the traditional AAE and our Hybrid AAE were trained on the MNIST dataset over 10 independent runs. During training, Gaussian noise was added to the latent vectors of both models to assess the robustness of the Hybrid AAE. Experimental results show that the Hybrid AAE significantly outperforms the baseline, achieving an average Frechet Inception Distance (FID) of 0.09 compared to 0.37 for the regular AAE. These findings validate our hypothesis that denoising the latent space using a Diffusion model improves the quality of generated samples.

1 Introduction

Generative models have become central to modern machine learning applications, enabling the synthesis of realistic data in a variety of domains. Among these, Adversarial Autoencoders (AAEs) stand out by combining the reconstruction capabilities of autoencoders with the distribution-shaping power of adversarial training [1]. Despite their success, AAEs are susceptible to noise in the latent space, which can negatively affect sample generation quality. In this paper, we introduce a Hybrid AAE that addresses this limitation by incorporating a Diffusion-based denoising model into the latent pipeline. Our goal is to empirically test whether such an integration leads to more robust and higher-quality sample generation.

2 Research Question

Can introducing a Diffusion model to denoise latent representations in an Adversarial Autoencoder improve the quality of generated samples in the presence of latent noise?

3 Background

3.1 Adversarial Autoencoders

Adversarial Autoencoders (AAEs) are a type of generative model that combines autoencoders with generative adversarial networks (GANs). The core idea is to enforce a prior distribution on the latent space using adversarial training. The encoder maps input data into a latent space, which is then fed into a decoder for reconstruction. Meanwhile, a discriminator tries to distinguish whether the latent codes come from the encoder or from a prior distribution such as a Gaussian. The encoder is trained to fool the discriminator, encouraging the encoded latent vectors to match the desired distribution [1]. This structure allows AAEs to function as both reconstruction and generation models, but the model’s sensitivity to noise in the latent space remains a challenge.

3.2 Diffusion Models

Diffusion models are a recent class of generative models that produce data by reversing a gradual noising process. During training, these models learn to denoise inputs step-by-step, approximating the reverse of a diffusion (Markov) process. The Denoising Diffusion Probabilistic Model (DDPM), introduced by Ho et al., demonstrated that a neural network could model the denoising steps effectively and generate high-quality samples from pure noise [2]. While these models are usually applied in the pixel space for tasks like image generation, their potential for denoising latent representations has not been widely explored. Our proposed Hybrid AAE applies a small diffusion model in latent space to clean up noisy encoded vectors before they are decoded.

3.3 Frechet Inception Distance (FID)

The Frechet Inception Distance (FID) is a widely-used metric to evaluate the quality of samples generated by generative models. It compares the distributions of real and generated images by modeling them as multivariate Gaussians in the feature space of an Inception-v3 network. Mathematically, the FID between two Gaussians with means μ_r , μ_g and covariances Σ_r , Σ_g is given by:

$$\text{FID} = \|\mu_r - \mu_g\|^2 + \text{Tr}\left(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2}\right)$$

Lower FID scores correspond to closer similarity between the real and generated data distributions, with scores below 10 often indicating high-quality generation in complex datasets [3]. In this work, we use FID to quantitatively compare the outputs of our regular and Hybrid AAEs.

4 Methods

4.1 Model Architectures

The architectures used in our experiments are as follows:

- **Encoder:** 784 (input) $\rightarrow$ 512 $\rightarrow$ 256 (latent)
- **Decoder:** 256 (latent) $\rightarrow$ 512 $\rightarrow$ 784 (output)
- **Discriminator:** 256 (latent) $\rightarrow$ 256 $\rightarrow$ 1
- **Diffusion Module (Hybrid only):** 256 $\rightarrow$ 256 $\rightarrow$ 256 $\rightarrow$ 256

All layers are fully connected with ReLU activations.

4.2 Training Setup

- **Dataset:** MNIST (handwritten digits)
- **Train/Test Split:** 60,000/10,000
- **Epochs:** 20 per run
- **Runs:** 10 independent runs for each model
- **Noise Injection:** Gaussian noise sampled from $\mathcal{N}(0, 1)$ added to latent vectors during training

4.3 Loss Functions

- **Autoencoder:** L1 Loss for reconstruction
- **Discriminator:** Binary Cross-Entropy for adversarial training
- **Diffusion Model:** Mean Squared Error (MSE)

4.4 Evaluation Metric

The quality of generated samples was evaluated using the Frechet Inception Distance (FID) [3]. Lower FID values indicate higher visual fidelity and diversity in generated samples.

5 Results

The average FID scores obtained over 10 runs for both model variants are shown in Table 1. Additionally, Figure 1 visualizes the FID scores across all 10 runs for both the Regular AAE and the Hybrid AAE. Figure 2 shows a sample images generated by the Regular AAE (using a seed value of 42) and Figure 3 shows a sample of images generated by the Hybrid AAE using the same seed value.

Model	Average FID
Regular AAE	0.37
Hybrid AAE (with Diffusion)	0.09

Table 1: FID scores averaged over 10 runs on MNIST

The Hybrid AAE consistently outperformed the regular AAE, achieving substantially lower FID scores across all 10 runs. This trend is clearly visible in the bar chart and supports our hypothesis that denoising the latent space leads to improved sample quality.

6 Future Work

Future research can explore the following extensions:

- **Architectural Improvements:** Testing convolutional or transformer-based Diffusion networks in latent space.
- **Dataset Generalization:** Applying the Hybrid AAE to more complex datasets such as CIFAR-10 and CelebA.
- **Latent Dimensionality:** Investigating how varying the size of the latent space affects denoising performance.
- **Joint Training:** Exploring end-to-end training of the encoder, diffusion module, and decoder.

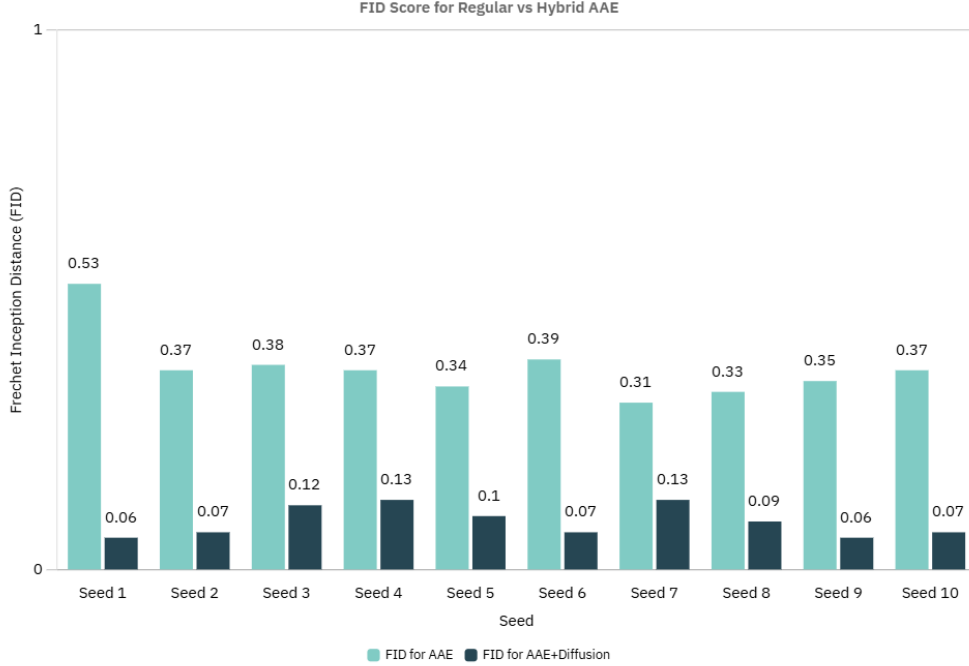


Figure 1: FID scores for Regular AAE (light green bars on the left) and Hybrid AAE (dark green bars on the right) across 10 training runs. Lower scores indicate better generation quality, and Hybrid AAE consistently achieves lower FID scores.

7 Conclusion

This paper introduces a Hybrid Adversarial Autoencoder architecture that incorporates a Diffusion model to denoise latent representations. By addressing noise in the latent space, the Hybrid AAE significantly improves the quality of generated samples, as evidenced by lower FID scores on the MNIST dataset. This work highlights the effectiveness of combining diffusion-based denoising with adversarial training and opens new directions for enhancing generative models.

References

- [1] A. Makhzani, J. Shlens, N. Jaitly, I. Goodfellow, and B. Frey, “Adversarial Autoencoders,” *arXiv preprint arXiv:1511.05644*, 2015.
- [2] J. Ho, A. Jain, and P. Abbeel, “Denoising Diffusion Probabilistic Models,” *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- [3] M. Heusel, H. Ramsauer, T. Unterthiner, B. Nessler, and S. Hochreiter, “GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium,” *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

Regular AAE Samples for Seed 42



Figure 2: A sample of images generated by the baseline AAE using a seed value of 42.

Hybrid AAE + Diffusion Samples for Seed 42



Figure 3: A sample of images generated by the Hybrid AAE which uses Diffusion, using a seed value of 42.